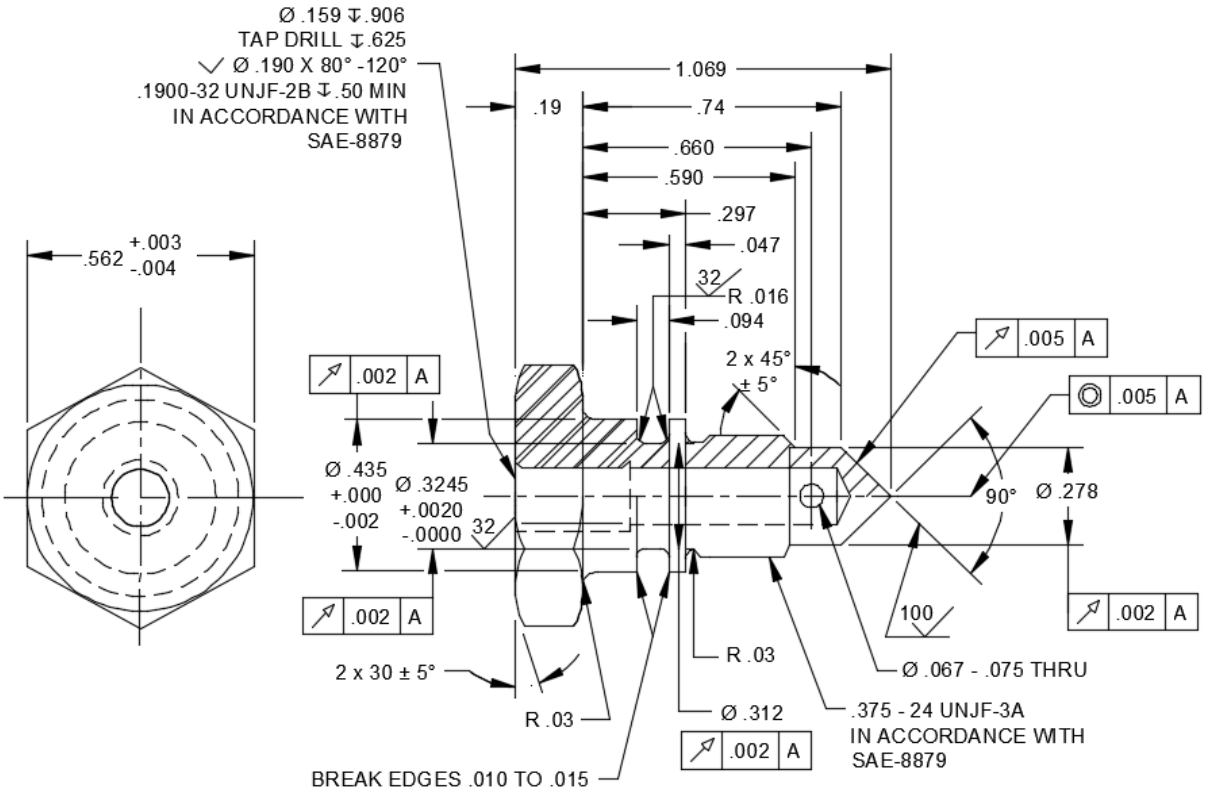
INCH-POUND
MS27611H
29 January 2018
SUPERSEDING
MS27611G
23 September 2015

DETAIL SPECIFICATION SHEET

VALVE, HYDRAULIC BLEEDER, AIRCRAFT WHEEL BRAKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.



NOTES:
1. Dimensioning and tolerancing: ASME Y14.5M.
2. Dimensions are in inches.
3. Unless otherwise specified, tolerances for two place decimals ±.010, three place decimals ± .005, and angles ± .5°.
4.. Break edges .003 to .015 unless otherwise specified.
5. Surface texture: Symbols in accordance with ASME Y14.36. Requirements in accordance with ASME B46.1. Unless otherwise specified surfaces to be 150 µin Ra.

FIGURE 1. Valve.

AMSC N/A

FSC 4820

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Requirements

Hydraulic bleeder valves shall be as specified on figure 1 and in table I.

Material: Steel rod, in accordance with the following:

Alloy 4130 in accordance with SAE-AMS-S-6758
Alloy 4140 in accordance with SAE-AMS6349 or SAE-AMS6382
Alloy 4330 in accordance with SAE-AMS6427
Alloy 8630 in accordance with SAE-AMS6280
Alloy 8740 in accordance with SAE-AMS6327

Heat-treat to C28RH to C33RH.

The valve shall withstand and seal 3,000-PSI min pressure.

Order O-Rings separately: Use O-ring packing in accordance with SAE-AS83461/1, part number M83461/1-011, or equivalent.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

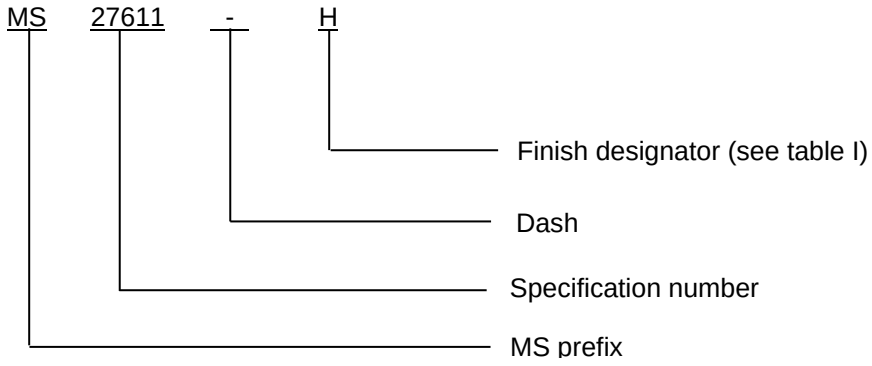
TABLE I. Finish identification codes.

Dash PIN code plating finish	Plating finish
Blank	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 3. 1/
-H	Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/
-J	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/

1/ Embrittlement test need not be run. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Hexavalent chromium free. Finish shall be ROHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.

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PIN: The PIN consists of the letters “MS”, the specification number, a dash and a letter for material finish designator.



PIN examples:
MS27611 indicates a hydraulic bleeder valve steel cadmium plated.
MS27611-H indicates a hydraulic bleeder valve steel zinc-aluminum plated.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Class I and II ozone depleting substances (ODS) shall not be used in MS27611 or any referenced procedures.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Marking: Finishes other than cadmium plating shall be impression stamped with finish designator H or J as applicable and may include the manufacturers CAGE, name, or trademark on the hex side only.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. This document references the following:

ASME-B46.1	SAE-AMS-6327	T9070-AL-DPC-020/077-2
ASME-Y14.5M	SAE-AMS-6349	
ASME-Y14.36	SAE-AMS-6382	
ASTM B117	SAE-AMS-6427	
ASTM F1136/F1136M	SAE-AMS-QQ-P-416	
SAE-AMS-C-81562	SAE-AMS-S-6758	
SAE-AMS2417	SAE-AS83461/1	
SAE-AMS-6280	SAE-AS8879	

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4820-2018-003)

Review activities:
Army - AT
Navy - SA, MC
Air Force - 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online data at https://assist.dla.mil.

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